Finecare™ FIA Meter Plus

Operation Manual

Model No.: FS-113

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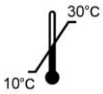
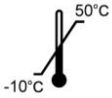
Index of Symbols

Read all contents of this Manual carefully prior to use.

⚠ **If the instrument is used in a manner not specified by the manufacturer, the protection provided by the instrument may be impaired.**

The following symbols are used on the **Finecare™ FIA Meter Plus**, related components and accessories, labels or in the text of this user manual:

	Biohazard
	Caution
	Consult Operator's Manual
	WEEE (Waste Electrical and Electronic Equipment)
	Keep Dry
	**Operation Temperature Limitation**
	**Storage and Transportation Temperature Limitation**
	Keep away from Sunlight
	For *in vitro* Diagnostic Use only
	Manufacturing Date
	Serial Number

	Manufacturer
EC REP	**Authorized Representative**
	Direct current
	ON (Power)
	Off (Power)

Section Ⅰ Introduction and Principle

This manual contains the instructions for the operation of the **Finecare™ FIA Meter Plus** and for the sample testing procedures. For specific tests' procedure please refer to the Instruction for Use of specific products.

Finecare™ FIA Meter Plus manufactured by Guangzhou Wondfo Biotech Co., Ltd., is a portable instrument for fluorescence detection to quantify concentration of various kind of analytes in human blood or urine. The Test Device and instrument are for *in vitro* diagnostic use only. **Finecare™ FIA Meter Plus** is intended for professional use.

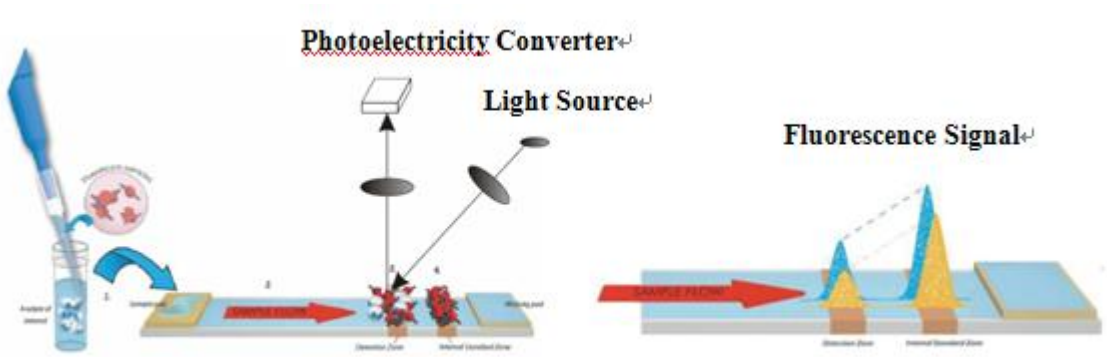
Finecare™ FIA Meter Plus uses an LED as the Excitation Light Source. The emitted light from the fluorescence dye is collected and converted into an electrical signal. The signals are closely related to the amount of fluoresceing dye molecules present on the spot under examination.

After a buffer-mixed sample is applied to the Test Device, the Test Device is inserted into **Finecare™ FIA Meter Plus** and the concentration of the analyte is calculated by a pre-programmed calibration process. **Finecare™ FIA Meter Plus** can only accept Test Devices that are designed specifically for use with this instrument.

Finecare™ FIA Meter Plus is equipped with a built-in Test Device Holder and does not require an external holder to place the Test Device in. The Test Device Holder will appear through the opening on the front surface of the instrument.

The Power ON/OFF Switch, located on the left of the instrument, which powers the **Finecare™ FIA Meter Plus.** The instrument has a built-in printer.

Figure 1 Working Principle



Section Ⅱ Composition of System

Unpacking

After opening the package box, compare the contents with the list below.

If any item is missing, please contact your sales distributor or **Guangzhou Wondfo Biotech Co., Ltd.** Detailed contact information is mentioned in ***Section Ⅺ***

Contents

1. Standard contents:

No.	Name	Quantity
1	Finecare™ FIA Meter Plus	1
2	Power adapter for instrument (including the power cord)	1
3	Quality Control Card	1
4	Ethernet cable	1
5	Printer paper (roll)	1
6	Operation manual	1
7	Quick operation instruction for use	1
8	Warranty Card	1
9	Packing List	1

Fig. 2.1 Structure Chart of Finecare™ FIA Meter Plus- Front View

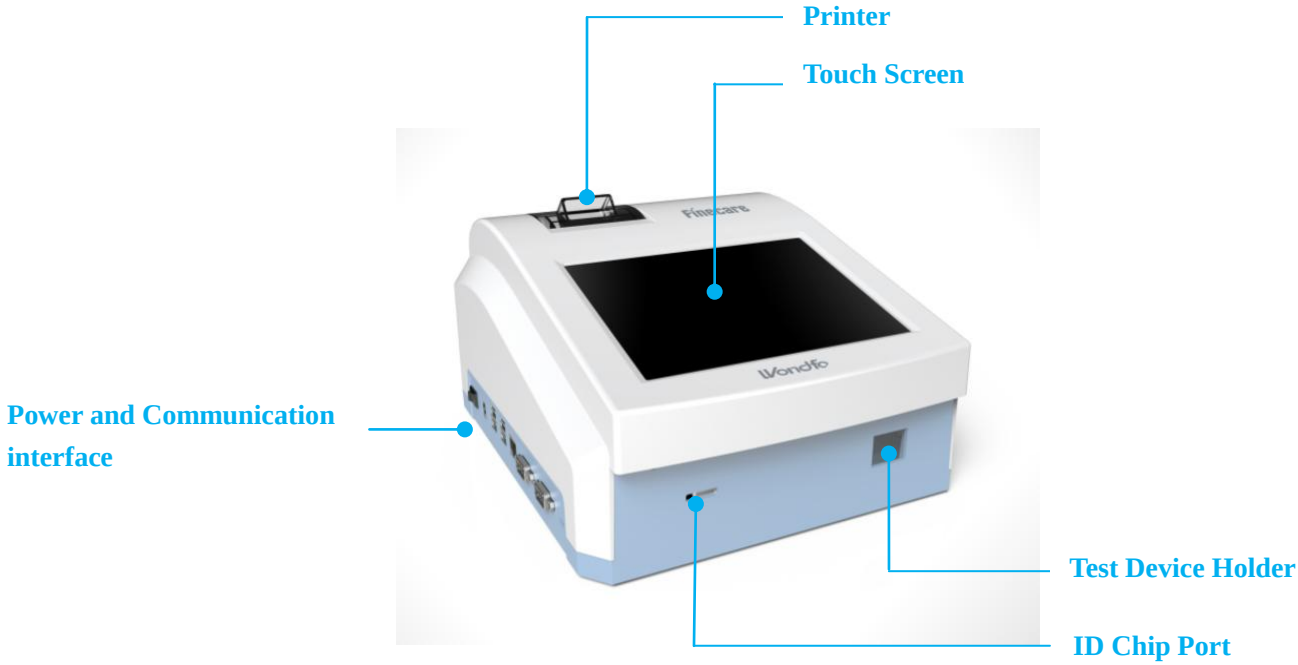


Fig. 2.2 Finecare™ FIA Meter Plus Features - Left View

Power switch, Ethernet Port, USB Interface, Serial Interface, Power interface, Selector switch for operating mode, which are used in **Finecare™ FIA Meter Plus**'s startup, shutdown, select the operating mode and connection with PC and Ethernet.

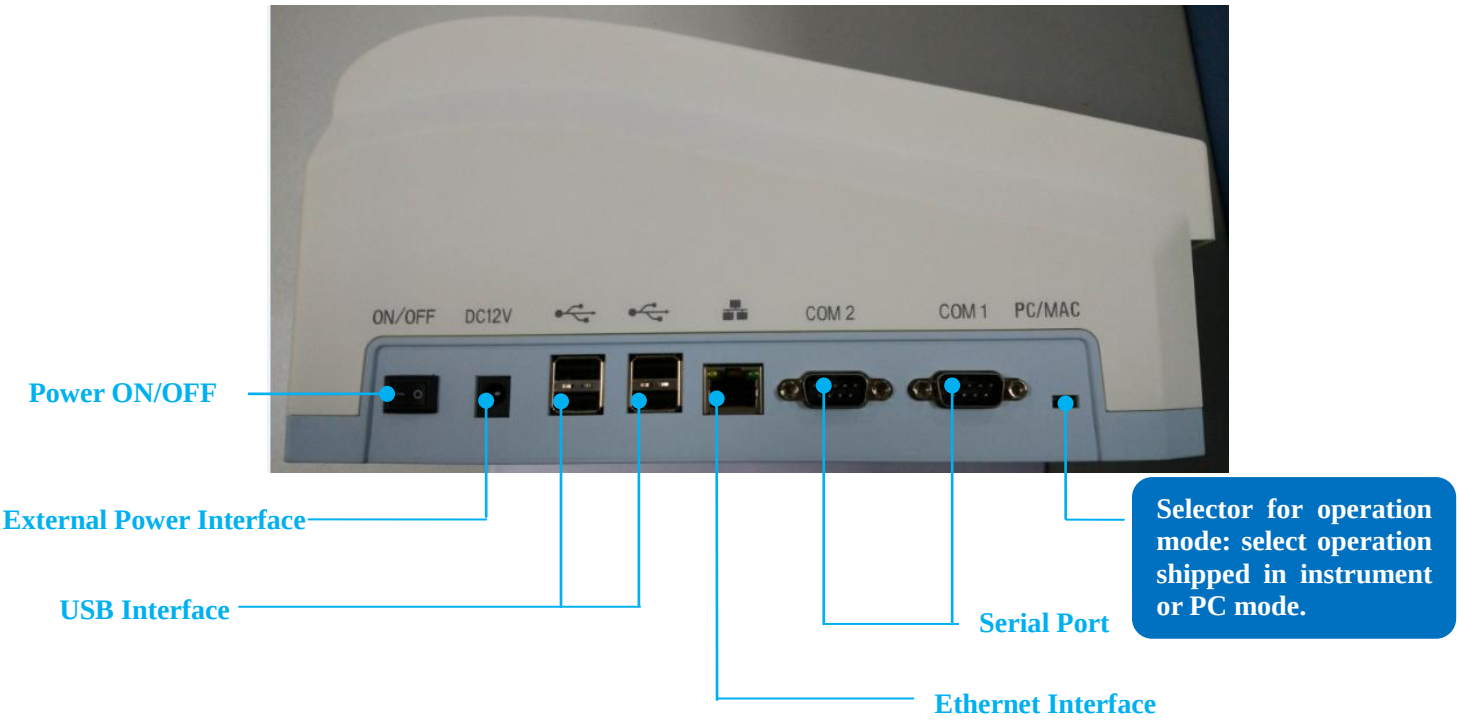
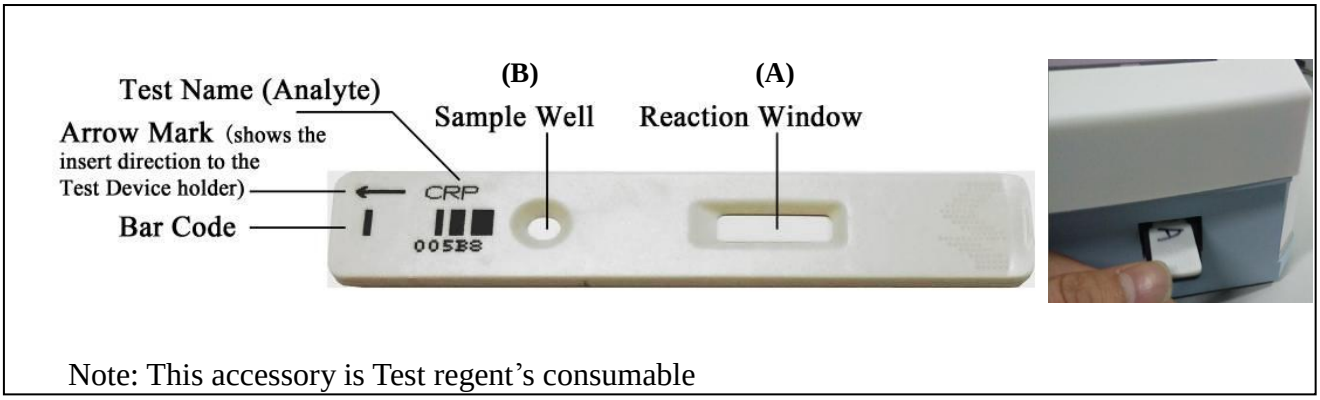


Fig. 2.3 Power Adapter

Consumables of Reagent Kit

Fig. 2.4 Test Device Features

(Remark: You shall buy the Test Devices separately)



Note: This accessory is Test regent's consumable

Fig. 2.5 Test Device ID Chip

(The ID Chip contains the information of test item and lot number)

Note: This accessory is Test regent's consumable

Fig. 2.6 Quality Control Card

Section Ⅲ Specifications and Operation Conditions

1. Basic Specification

Model No.	FS-113
Excitation light source	LED
Excitation spectrum	Center wavelength λ_0 = 470nm
Absorption spectrum	Center wavelength λ_1 = 525nm
Test Specimen	Whole blood, serum / plasma and urine
Software system	English and Chinese input method, customized intelligent management module
Interface	RS232, USB, Ethernet network
Built-in Printer	Thermal
Size	270*238*146mm
Weight	About 2800g

Operating Environment

Temperature		10℃~30℃（50℉－86℉）
Relative Humidity		10%~80%
Atmospheric pressure		86kPa~106kPa
Power Supply	Adapter	Input: 100-240Vac, 50-60Hz, 1.4-0.7A Output: 12V⎓ 5.0A 60W MAX
	Power Input:	12V ⎓，45W
Location		Dry, clean, flat, horizontal surface away from direct sunlight, mechanical vibration, and strong electromagnetic interferences.

Storage and Transportation Environment

Temperature	-10℃~+50℃
Relative Humidity	≤85%

Performance

Repeatability	CV ≤ 15%
Stability	CV ≤ 8%

Section Ⅳ Initial Installation Procedure

1. Power Requirements

If you are uncertain of your power line, consult with your utility manager. The **Finecare™ FIA Meter Plus** can only be used with the power adapter (including the power cord) provided. If the replacement for Power Adaptor is required, please consult with your sales distributor or Guangzhou Wondfo Biotech Co., Ltd.

2. Installation

Please be noted that **Finecare™ FIA Meter Plus** should only be used in 10℃ ~ 30℃ (The operational temperature of Test Devices should refer to relevant Instruction for Use) and Relative Humidity of 10% ~ 80% conditions (Detailed information is mentioned in ***Section Ⅲ***).

①. Place **Finecare™ FIA Meter Plus** on a dry, clean, flat, horizontal surface, with the front surface at least 10 cm inside of the table edge.

②. Connect the **Power Cord** with **Finecare™ FIA Meter Plus** to the wall power outlet.
Note: *The meter shall be connected to a mains socket outlet with a protective earthing connection. The mains outlet shall be located in a user accessible position for easy unplugging in emergency. The mains plug of the power cord set is a disconnect device of the instrument. In order to completely disconnect the instrument from the supply mains, the mains plug of the power cord set should be unplugged from the mains outlet.*

③. Turn on the **Power ON/OFF Switch** on the left of **Finecare™ FIA Meter Plus t**o carry out power startup and shutdown.(See **Fig. 4. 1)**

Fig. 4.1

Power ON/OFF

External Power Interface

Selector for operation mode

Caution:
*Keep the power cord for the **Power Adapter** in suitable place. Power cord without good maintenance could render hazardous to connect/disconnect the **Power Adapter**.*

Section Ⅴ Operation Instruction

Finecare™ FIA Meter Plus provides two test modes, including **Standard Test** and **Quick Test.** Under the standard test mode, the instrument as a incubation place reaction of test item to provide required standard countdown. Under the quick test mode, the test item shoud be completed incubation prior, the user can directly conduct testing.

Finecare™ FIA Meter Plus can be operated through LCD screen with finger touch or touch pen, also can be operated through with a mouse and keyboard.

V.1 Start

V.1.1 Turn on the **Power ON/OFF** on the left of **Finecare™ FIA Meter Plus.**

Fig 5.1

V.1.2 The instrument will load the test system.

V.1.3 Waiting for 1 minute, the instrument will show the initial screen, and slot pop-up the Test Device holder automatically.

V.1.4 **Quick Test** interface will be showed by default when you turn on **Finecare™ FIA Meter Plus** at the first time, after then boot screen displays the initial screen that the user once left off. (See **Fig. 5.2**)

Fig 5.2 Quick Test interface

A **Function Buttons** - Touch the button to achieve the corresponding function.
B **Information Input** - Input Serial Number, Sample number and Sample type.
C **Results Display** - Display test serial number, item name and test results.
D **Quick Set** - Reset the slot, slot out, send test result to LIS system, print and print automatically.

①【Temperature Display】Display the temperature, the bald thermometer will turn to red by default when the temperature is above 30℃, the default temperature can be set in the system setting.
②【Test】To carry out testing of test item.
③【History】Inquire historical record.
④【Account】Account the test data.
⑤【QC】Finecare™ FIA Meter Plus selftesting of quality.
⑥【Set Up】Set instrument parameters.
⑦【Test Mode】Choose test mode.
⑧【Start Test】 Start to test.

Caution:
If the initial interface doesn't show after more than 3 minutes please turn off the ***Power ON/OFF Switch*** *and restart the instrument. If failure continues, please contact your sales distributor or Guangzhou Wondfo Biotech Co., Ltd.*

V.2 Test Operation Example

Please insert the ID chip of corrsponding test item before clicking "**Test**", Under the initial interface of **Finecare™ FIA Meter Plus**, the user can operate instrument by clicking the corresponding function button for invoking different procedures.

Finecare™ FIA Meter Plus enter the initial test interface after booting, you can freely choose **"Standard Test"** mode or **"Quick Test"** mode according to different needs. (See **Fig. 5.3**)

Fig. 5.3

Caution:

1. It is recommended to insert ID chip after successful boot!

*2. Please make sure the "***selector switch for operating mode"** *on the left side of the instrument to be appropriated for "MAC", when you choose the test mode of* **Finecare™ FIA Meter Plus'** *system.*

2.1 Choose Test mode (**Standard Test** or **Quick Test**).

2.2 Input basic patient medical information, including serial number, sample number and sample type. (The number will be automatically generated by system if you don't input it.) .Click" **Detailed information**" to input more detailed information.

2.3 Add sample to the buffer, mix well.

Caution:

Please carefully read relevant Instruction for Use of specific Test Device before adding sample to test, and operate the test according to requirements.

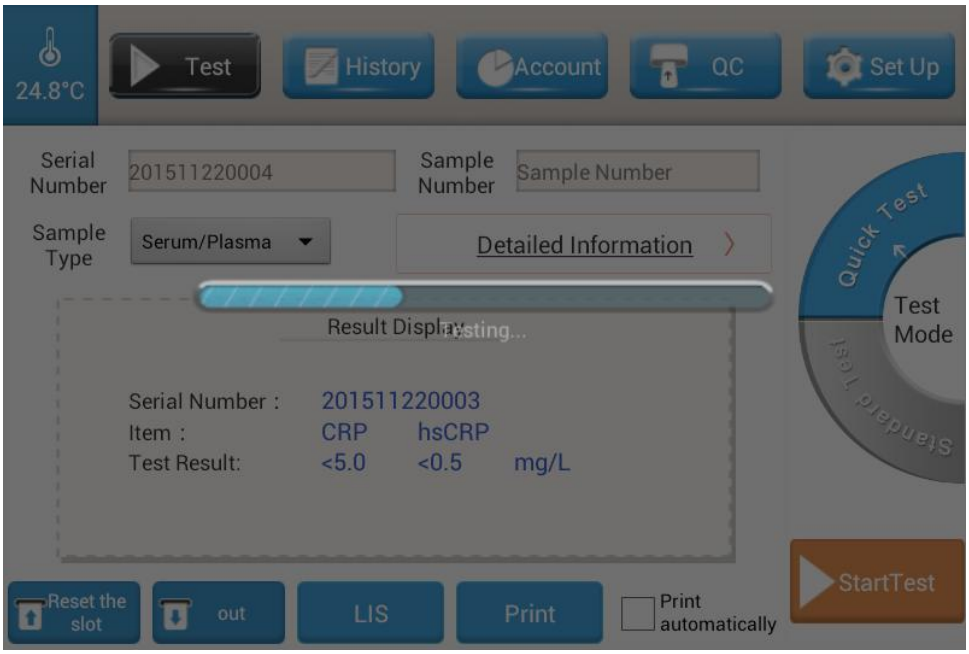
2.4 Prepare a Test Device and apply sample-buffer mixture onto the sample well of Test Device (Refer to the Instruction for Use of specific Test Device).

2.4.1 If choose **Quick Test** at step 2.1, insert the Test Device that is completely reacted into the Test Device Holder of **Finecare™ FIA Meter Plus**, and click "**StartTest**".

2.4.2 If choose **Standard Test** at step 2.1, immediately insert the Test Device that is into the Test Device Holder of **Finecare™ FIA Meter Plus**, and click "**StartTest**".

2.5 Finecare™ FIA Meter Plus will scan the Test Device automatically, the screen shows "**Testing**" and testing progress. (See **Fig. 5.4**)

Fig. 5.4



Caution:
*If the lot number of the **ID Chip** does not match with the lot number of the Test Device when use the specific Test Device at the first time, the instrument will show **"insert the corresponding ID chip"**. Please check the lot number of ID Chip and Test Device if they are mismatched, please click " **Change reagent and test again** " ,* Finecare™ FIA Meter Plus *will pop-up Test Device holder and show " **Cancel test** " on the screen. If the lot number of **ID Chip** and Test Device are matched, please click "**OK**",* Finecare™ FIA Meter PLUS *will automatically recheck ID chip and Test Device.*

2.6 The test result will be showed under the "Result Display" of the screen. (See **Fig. 5.5**)

Fig. 5.5

2.7 Click "**Print**" to print the test result.

Caution:
The test result will be printed automatically after ticking off "**Print automatically**".

2.8 Take out the Test Device. Insert the next Test Device that is completely reacted for the next testing.

2.9 Click "**Read ID card** ", **Finecare™ FIA Meter Plus** will read and keep the ID card information.

2.10 When **Finecare™ FIA Meter Plus** is not used, click "**Reset the slot**", then turn off the power switch and unplug the power source of the instrument.

2.11 Click "**Reset the slot**" after completing the test.

V.3 History Operation Example

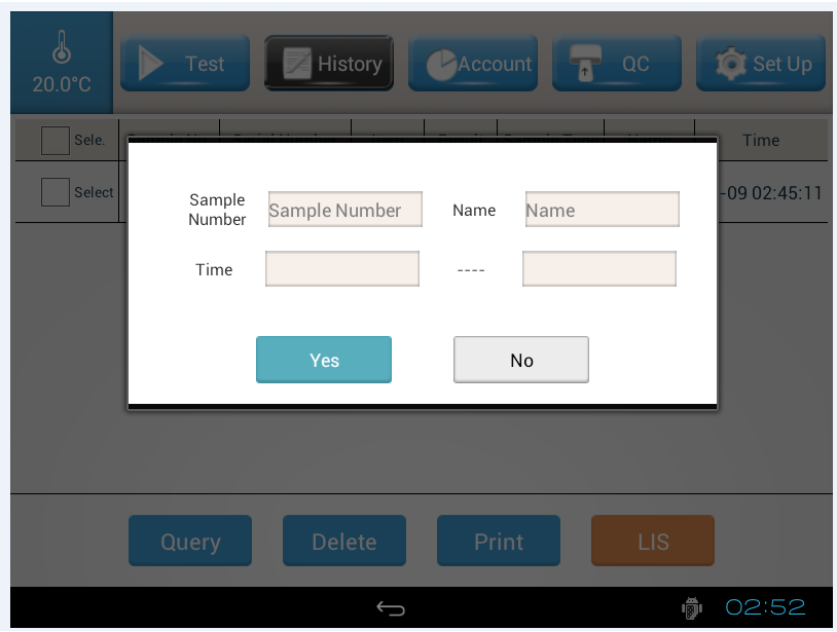
"**History**" function provides the user with a function of conditional filter, view the history of test records. There are Query, Delete and Print buttons to carry out operation of query, delete and print test results under the interface of "**History**".

3.1 Click "**History**" on the top of the instrument to enter the query interface, the test results of intraday will be listed. (See **Fig. 5.6**)

Fig. 5.6

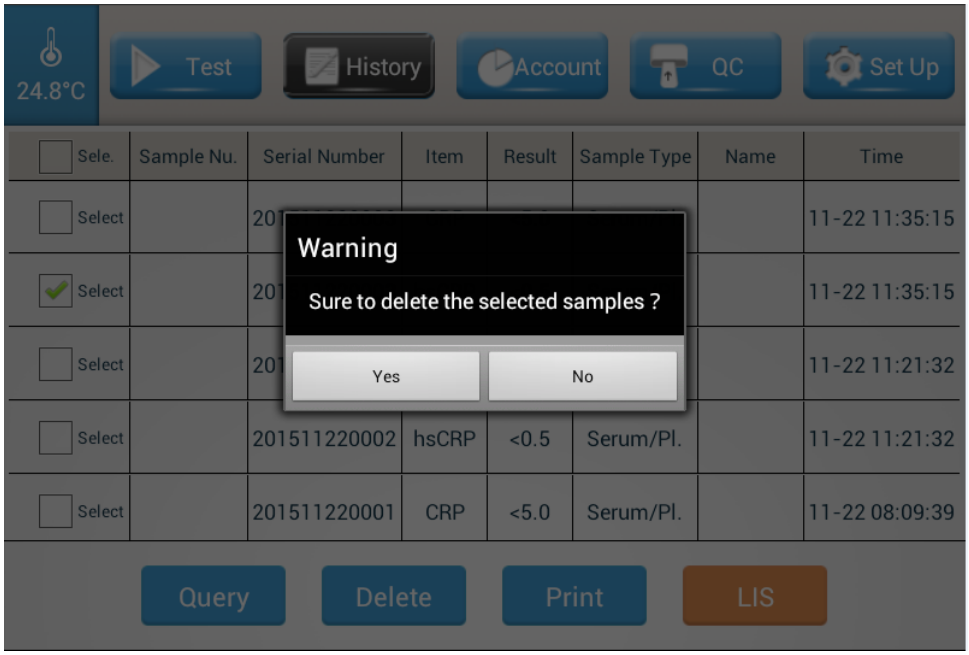
3.2 Click "**Query**" under the "**History**" interface, then input the query conditions, Finecare™ FIA Meter Plus will filter out the test results that match the query conditions to conduct fuzzy query. (See **Fig. 5.7**)

Fig. 5.7



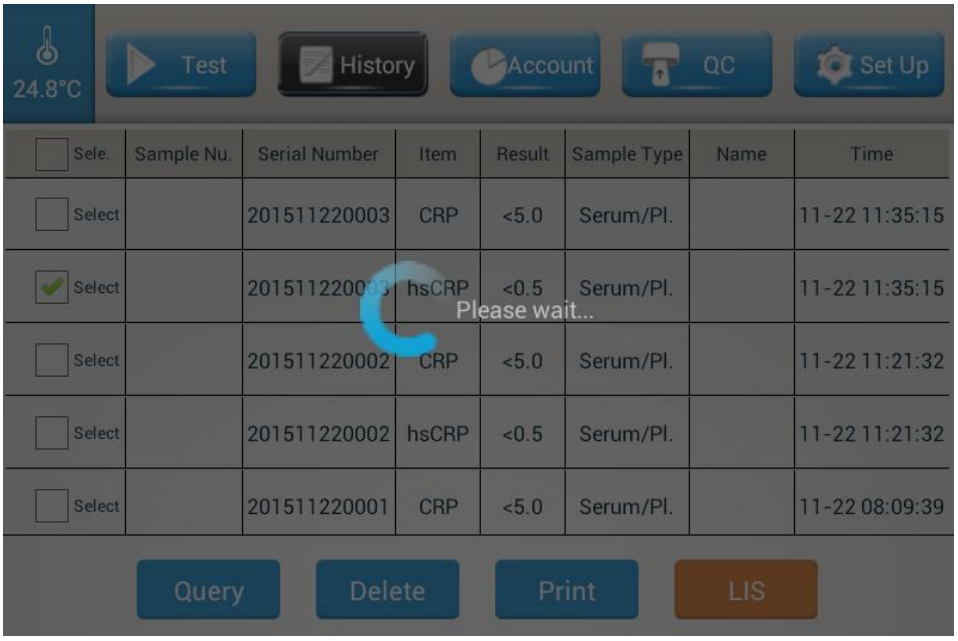
3.3 Tick all or specific data record, click "**Delete**" under the "**History**" interface, then click "**Yes**" to delete the selected records. (See **Fig. 5.8**)

Fig. 5.8



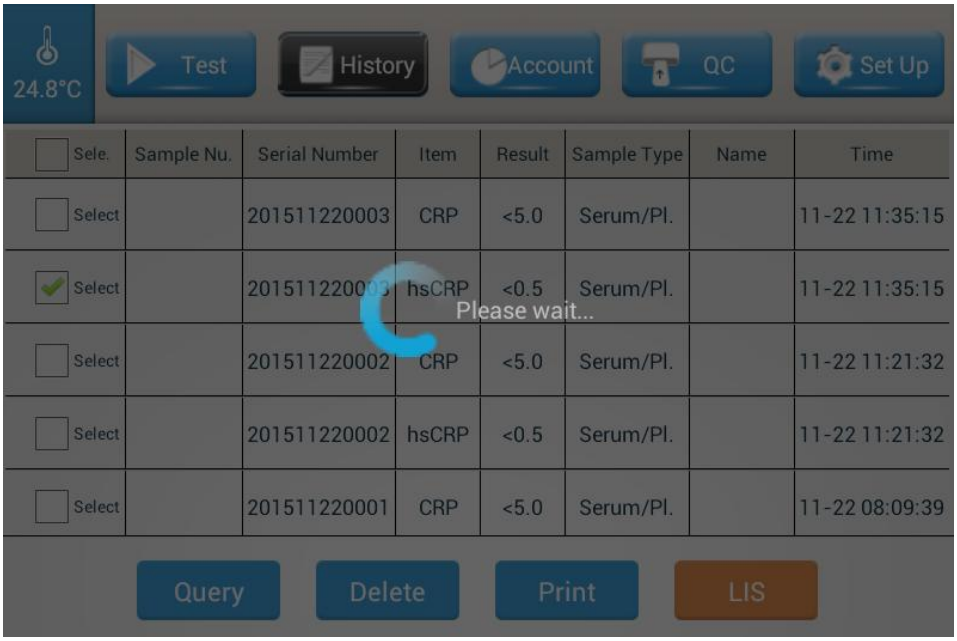
3.4 Tick all or specific data record, click "**Print**" under the "**History**" interface to print a selected records. When the selected data is greater than 10, only 10 data can be printed. (See **Fig. 5.9**)

Fig. 5.9



3.5 In the case of the LIS system was connected, select all or specific data record, click "**LIS**" under the "**History**" interface to send corresponding test result to LIS system. (See **Fig. 5.10**)

Fig. 5.10

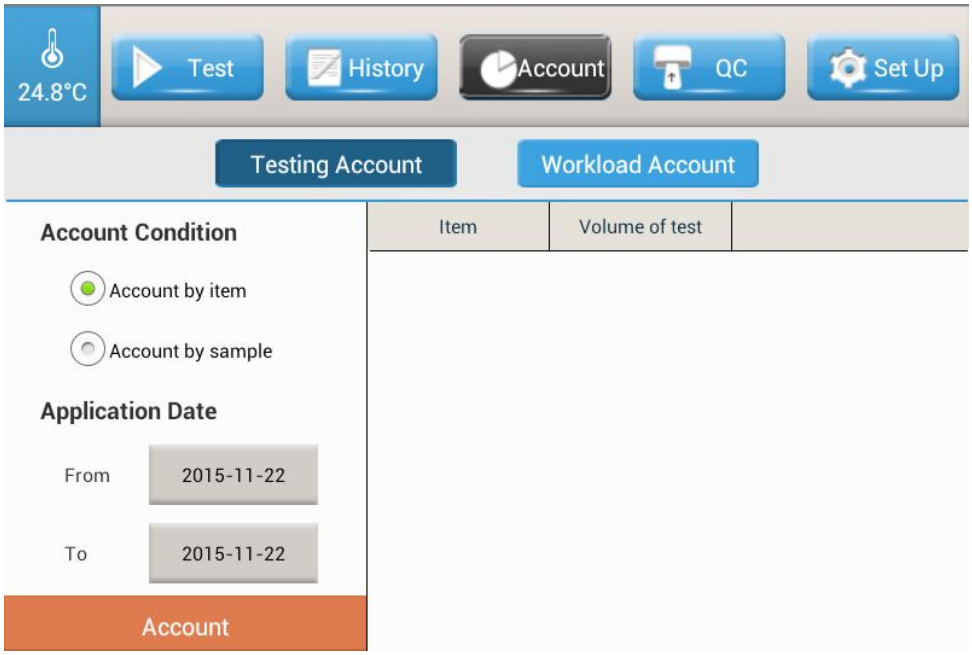


V.4 Account Operation Example

Click "**Account**" on the top of the instrument to enter the statistics interface. "**Account**" function provides the user with "**Testing Account**" and "**Workload Account**" function. Click "**Testing Account**" and "**Workload Account**" to show statistical results according to statistic conditions.

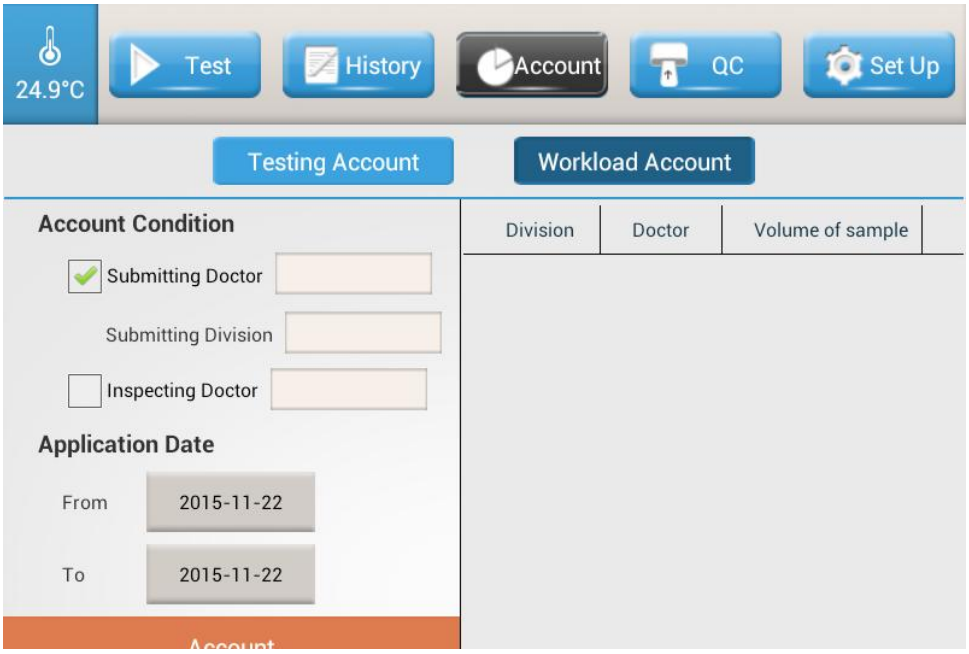
4.1 **Testing Account**: Count the number according to items or samples. (See **Fig. 5.11**)

Fig. 5.11



4.2 **Workload Account**: Workload can be **accounted** according to submitting doctor, submitting division and inspecting doctor. (See **Fig. 5.12**)

Fig. 5.12

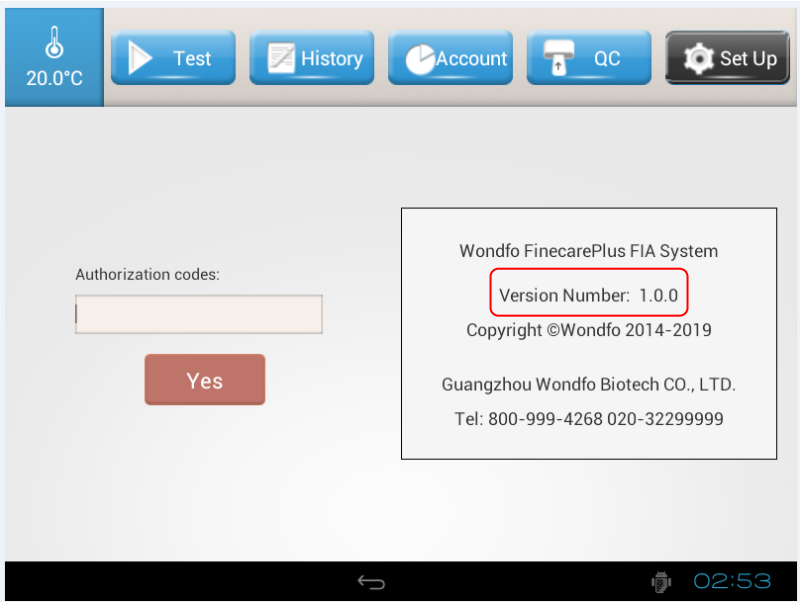


V.5 Set Up Operation Example

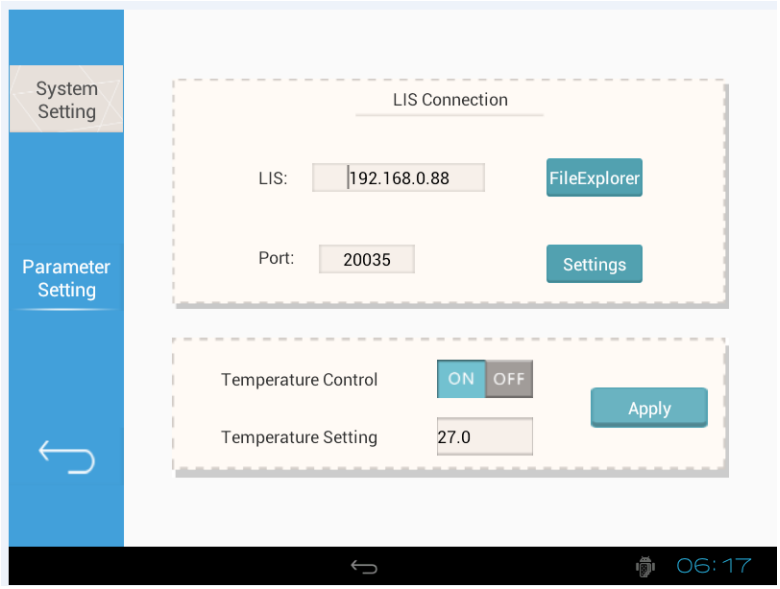
"**Set Up**" function provides the user with setting function for the machine function parameters, there are two mode for users, including general users and R & D users, they can enter the appropriate settings mode by using different password.

Click "**Set Up**" to see the Software version， input the password in the "**Authorization code** " field , then click "Yes" to enter the setting interface.

> ***Caution:***
> *The users* who no password permissions do not have permission about parameter settings.

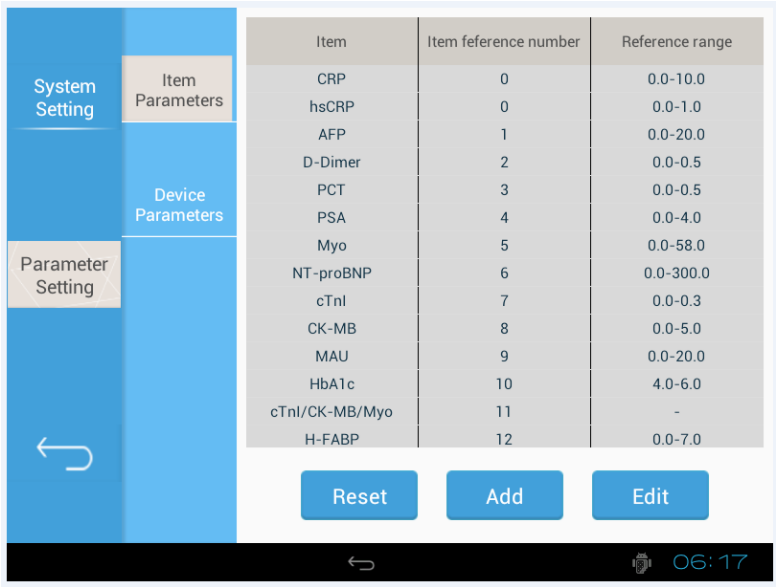


5.1 General user can only enter the "**System Setting**" interface to set LIS connection, bald temperature parameters.

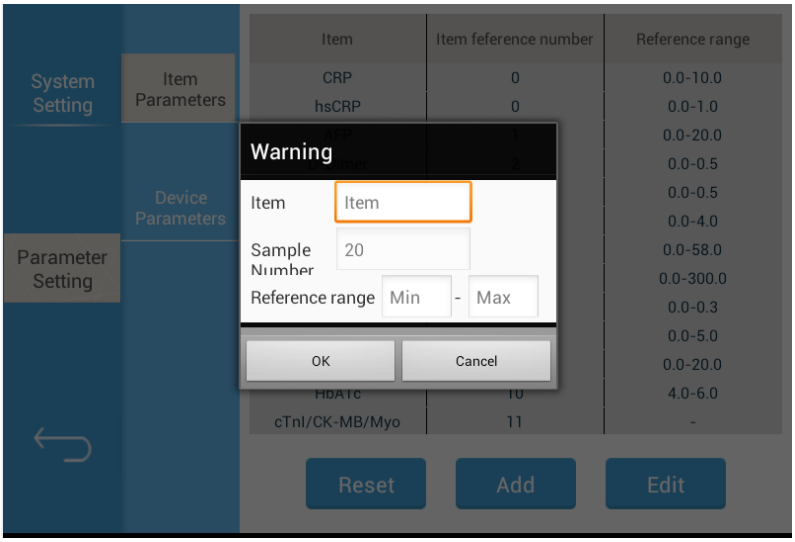


5.2 R & D users interface, in addition to the "**System Setting**" function, also increase the "**Parameter Setting**" function.

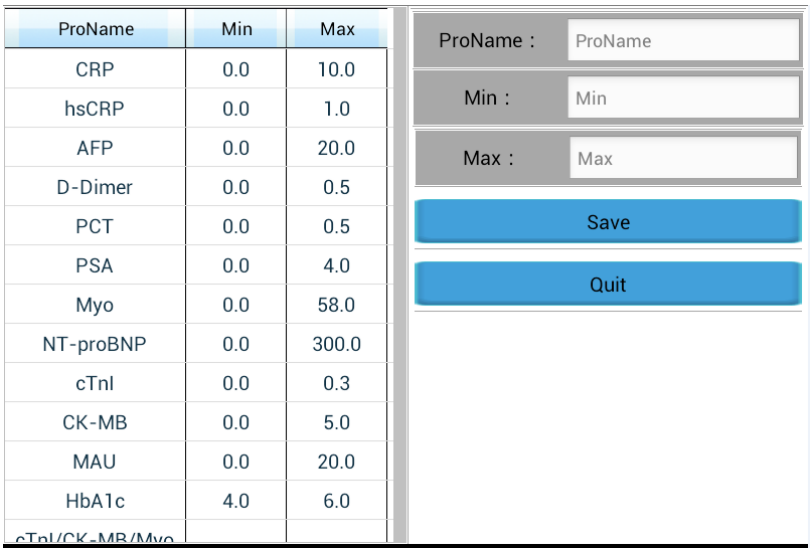
5.2.1 ”**Item Parameters**” under the "**Parameter Setting**" interface, the users can add and modify test items and the reference range.



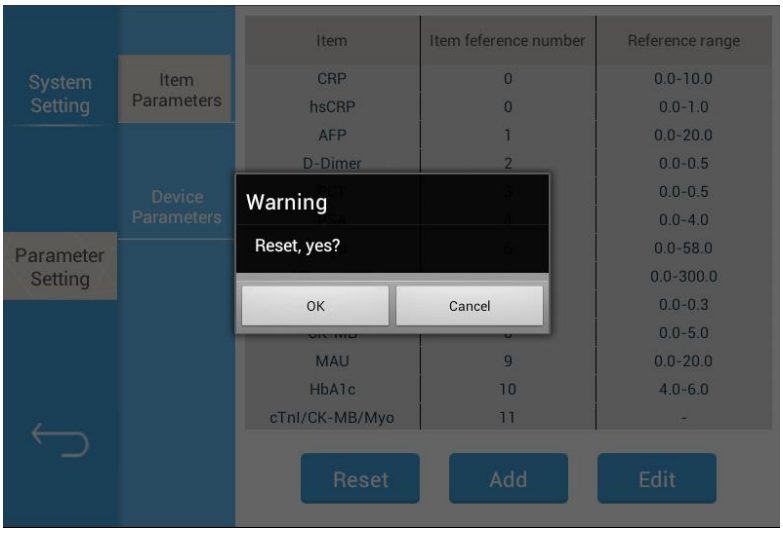
5.2.2 Click “**Add**” to add items.



5.2.3 Click “**Edit**” to modify the existing items’ parameters.



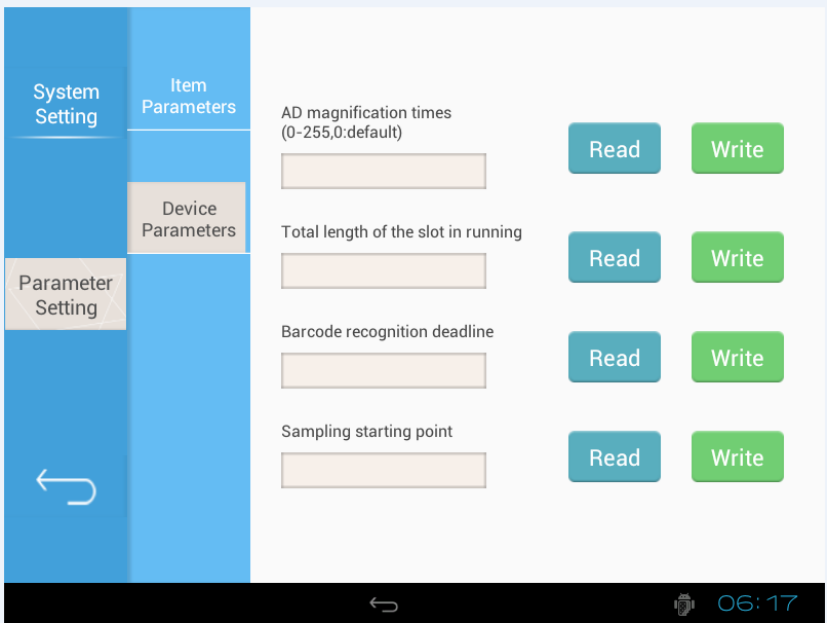
5.2.4 Click “**Reset**” to restore item parameters to the start parameter, then click ”OK” to complete the reset:



5.2.5 ”**Device Parameters**” under the "**Parameter Setting**" interface, the users can set “AD magnification times”, “Total length of the slot in running”, “Barcode recognition deadline” and

"Sampling starting point".

AD magnification times: Magnification of fluorescence signal.
Total length of the slot in running: The maximum distance of running forward when motor start with photoelectric switch.
Barcode recognition deadline: The timeout period of barcode recognition.
Sampling starting point: The starting position of fluorescence data collection.



Caution:
Please note that this setting must be executed by authorized technical service personnel, unauthorized users shall not operate.

Section Ⅵ Instruction for connection with LIS System and PC

This section introduce the operational instruction of **Finecare™ FIA Meter Plus** connect with LIS and PC.

Transfer Protocol Instruction:
Finecare™ FIA Meter Plus sends data to outside by UDP, click "**LIS**" when the test is completed every time, the current test data will be sent out.

Preset initial IP address of **Finecare™ FIA Meter Plus**: 192.168.0.3
Preset initial subnet mask of **Finecare™ FIA Meter Plus**: 255.255.255.0
Preset initial default gateway of **Finecare™ FIA Meter Plus**: 192.168.0.1
Preset initial receiving IP of PC: 192.168.0.88
Subnet mask of PC: 255.255.255.0
Default gateway of PC: 192.168.0.1
PC port: 20035
Communication mode: UDP

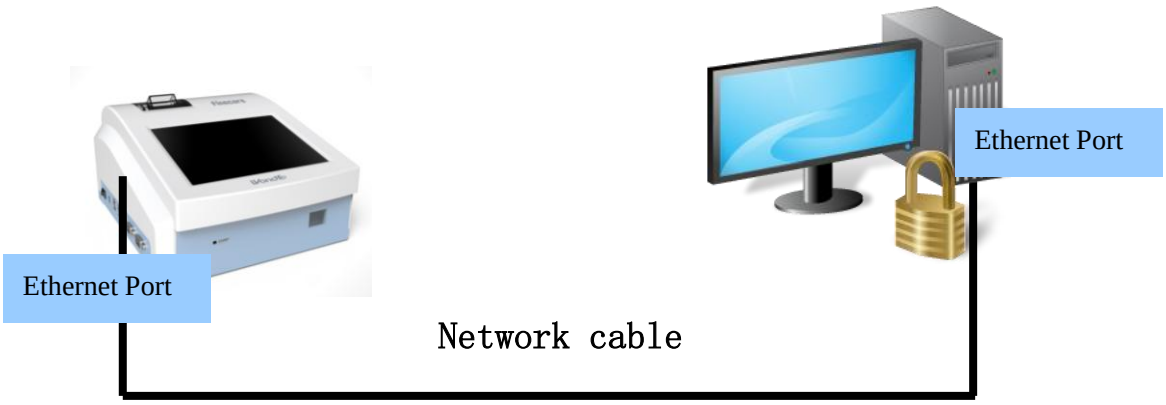
PC Data receiving format:
For CRP: FF&test serial number&CRP test result&hs CRP test result&test time&item name&EE
Example: FF&1&<0.5&0.93&2012-07-30 13:59:33& Whole Range CRP&EE

Ⅵ.1 Methods of Finecare™ FIA Meter Plus connect with LIS \ PC.

Method 1: When **Finecare™ FIA Meter Plus** connects to exclusive computer (see **Fig. 6.1**)

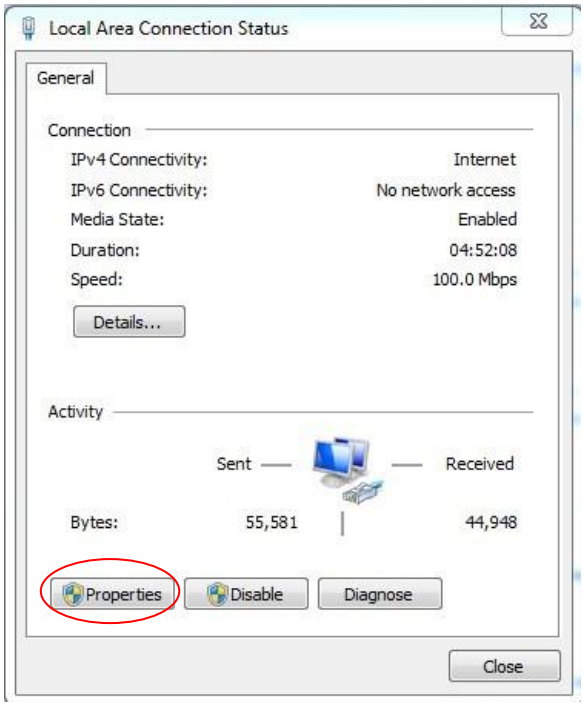
1) Use a standard network cable that can make normal PC access the Internet, one end of the network cable connects **Finecare™ FIA Meter Plus**'s Ethernet port, the other end connects the PC 's Ethernet port.

Fig. 6.1



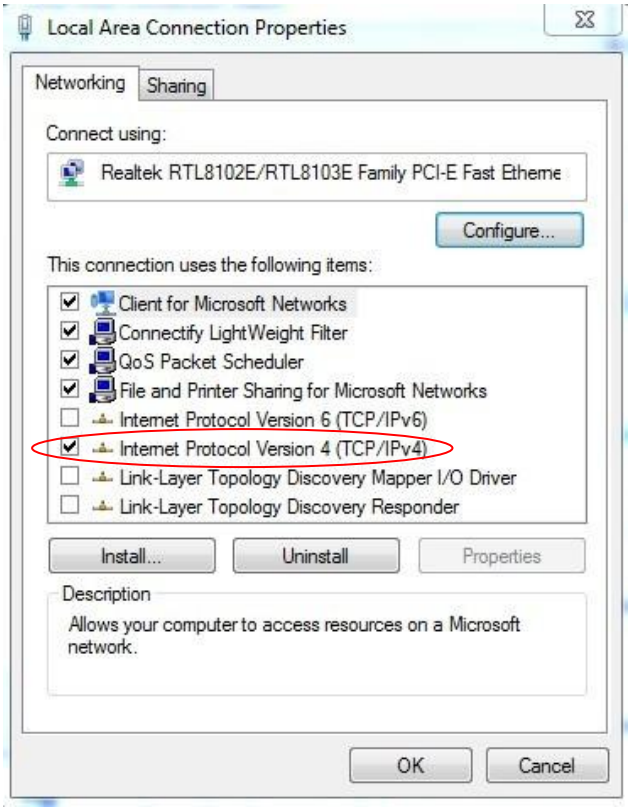
2) Set the IP address of PC according to protocol, click "Start" -> "Routine"-> "Attachment"-> "Communication"-> "Network"; double click "local connection" and enter to "Local Area Connection Status".(See **Fig. 6.2**)

Fig. 6.2



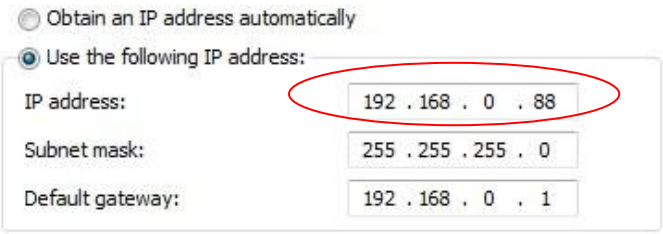
3) Click "Properties" and enter to "Local Area Connection Properties". (See **Fig. 6.3**)

Fig. 6.3



4) Double click "Internet Protocol Version 4 (TCP/IPv4)". Enter to "Internet Protocol (TCP/IP) Properties". Set IP address as 192.168.0.88， click "OK" after finishing. (See **Fig. 6.4**)

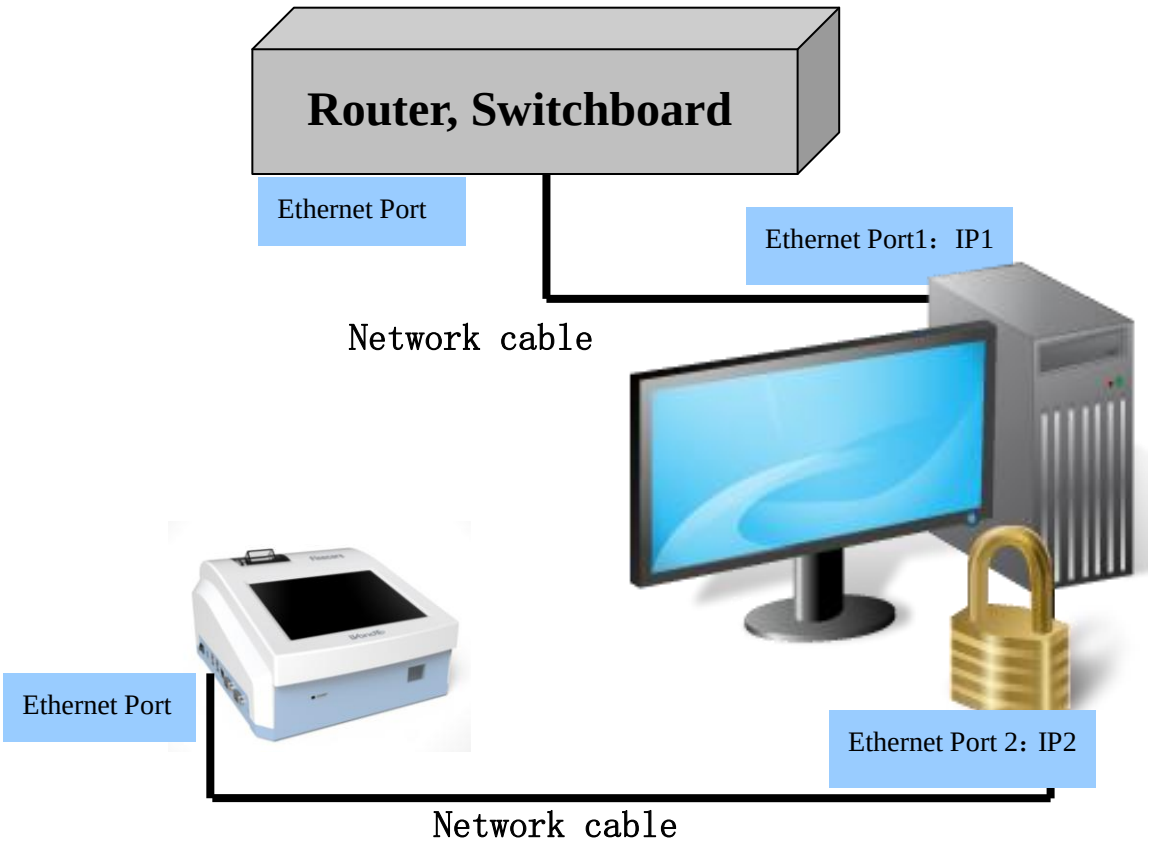
Fig. 6.4



Method 2: Multiple Network Information Card

Install one more NIC (Network Information Card) and corresponding driver in computer to work properly. Then use a standard network cable that can make normal PC access the Internet, one end of the network cable connects **Finecare™ FIA Meter Plus**'s Ethernet port, the other end connects PC 's Ethernet port of new Installed NIC. Alter the new NIC IP address in computer, the method as same as **Method 1**. (See **Fig. 6.5**)

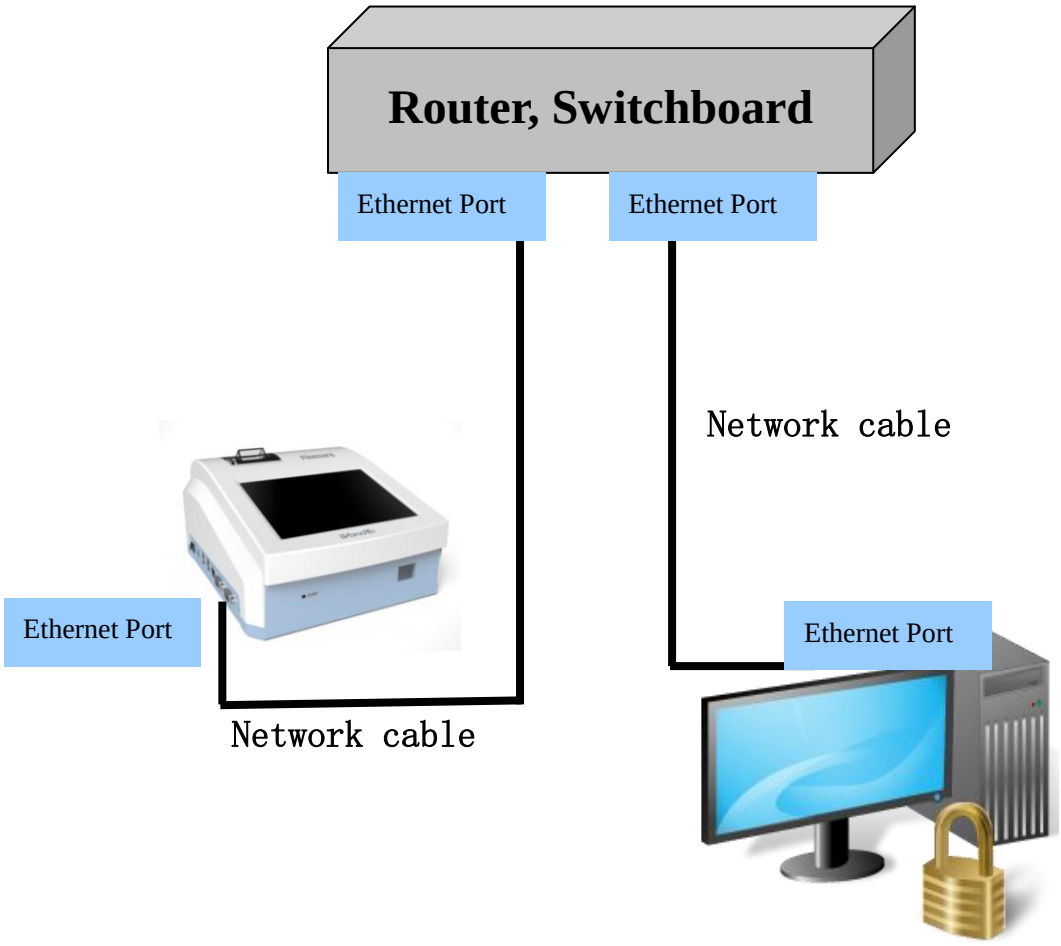
Fig. 6.5



Method 3: LAN (Local Area Network)

Use a standard network cable that can make normal PC access the Internet, one end of the network cable connects **Finecare™ FIA Meter Plus**'s Ethernet port, the other end connects Laboratory LAN(such as router and switchboard etc.). (See **Fig. 6.6**)

Fig. 6.6

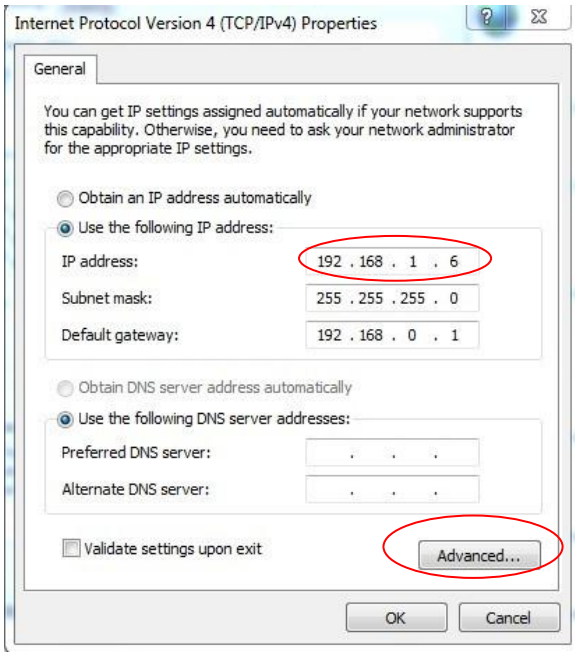


Click “Start” -> “Routine”-> “Attachment”-> “Communication”-> “Network”; double click “local connection” and enter to “Local Area Connection Status”, click “Properties” and enter to “Local Area Connection Properties”, double click “Internet Protocol Version 4 (TCP/IPv4)” and enter to “Internet Protocol Version 4 (TCP/IPv4) Properties”.

Don’t change primary IP address of PC. Example: Primary IP address: 192.168.1.6.

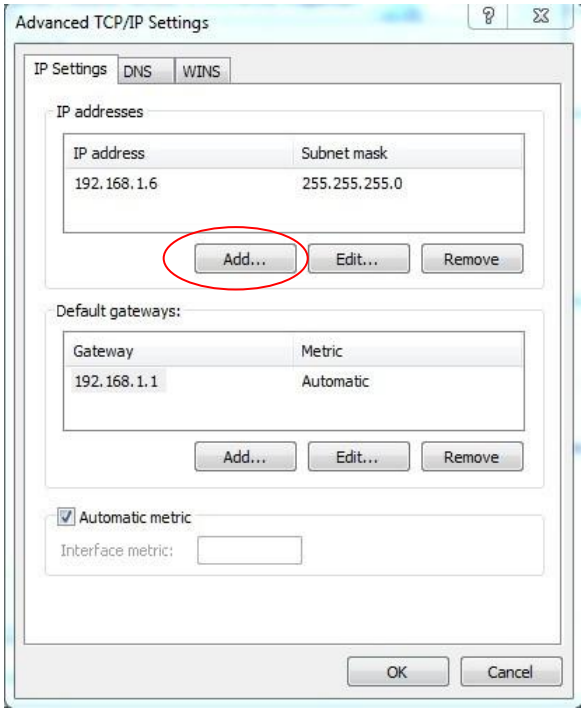
1) Cick “Advanced”, enter to “Advanced TCP/IP settings” (See **Fig. 6.7**)

Fig. 6.7



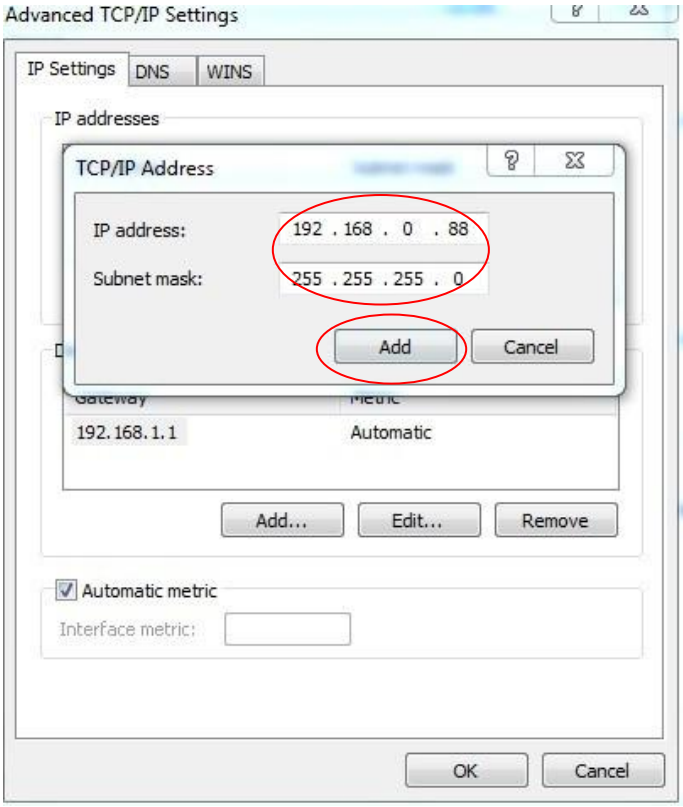
2）Click “Add” under “IP address”, enter to“TCP/IP address”. (See **Fig. 6.8**)

Fig. 6.8



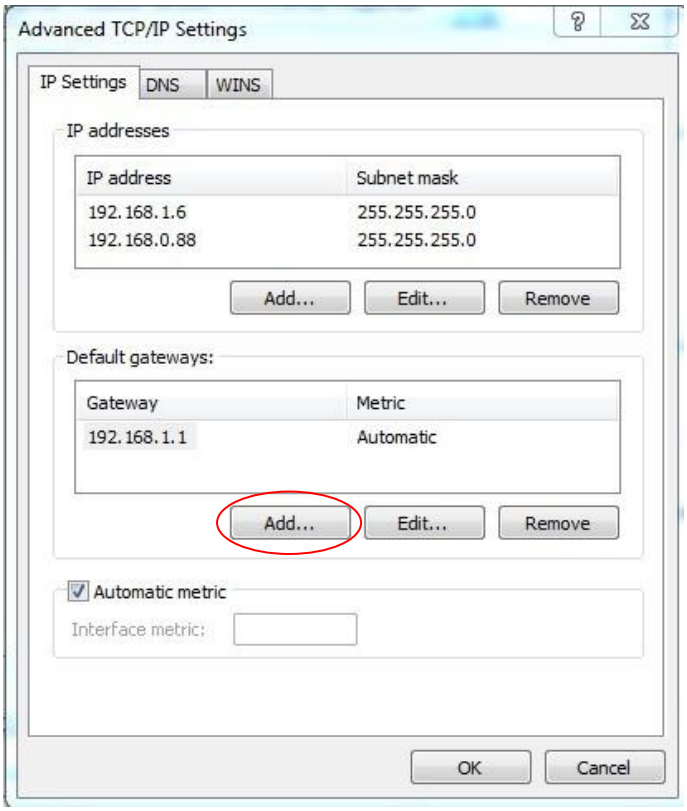
3）Add IP address：192.168.0.88，Subnet mask：255.255.255.0 (See **Fig. 6.9**)

Fig. 6.9



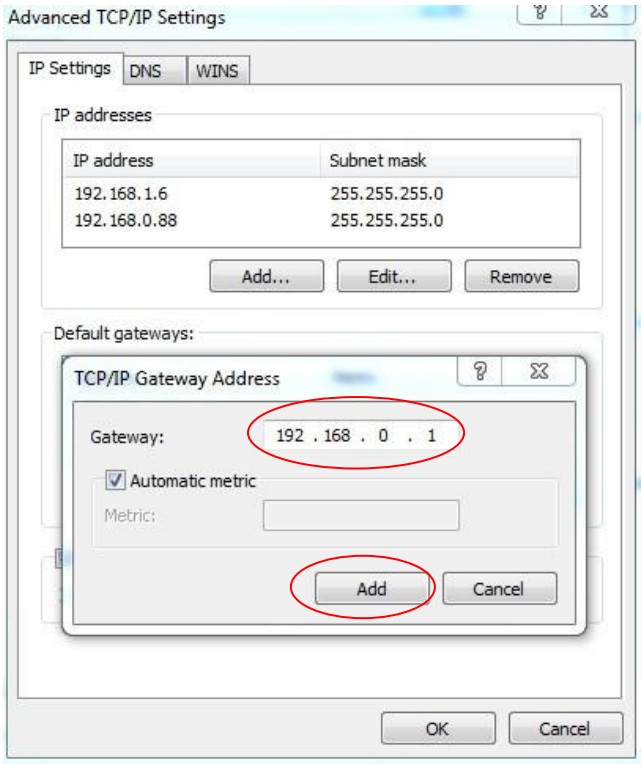
4）Then click “Add” to return to “Advanced TCP/IP settings” (See **Fig. 6.10**)

Fig. 6.10



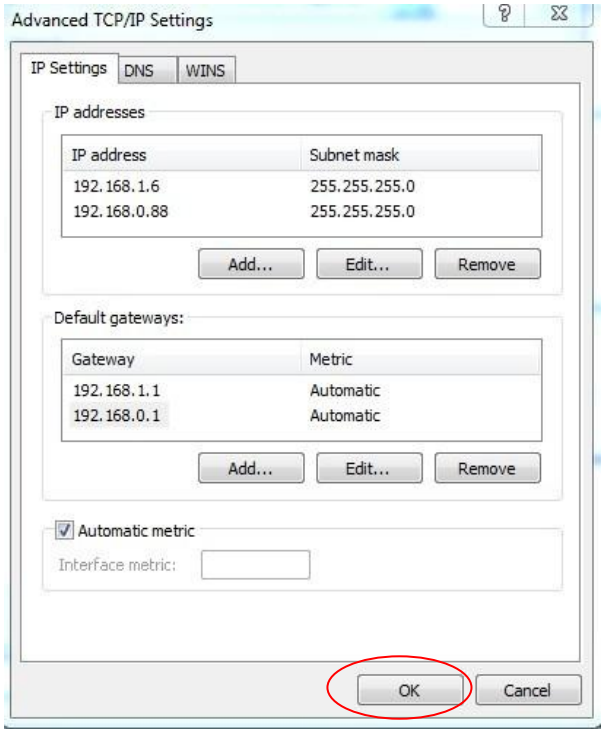
5）Click “Add” under “Default gateway”， enter to “TCP/IP Gateway Address”. (See **Fig. 6.11**)

Fig. 6.11



6）Input Gateway: 192.168.0.1, click “Add” after finishing, return to “Advanced TCP/IP settings”. (See **Fig. 6.12**)

Fig. 6.12

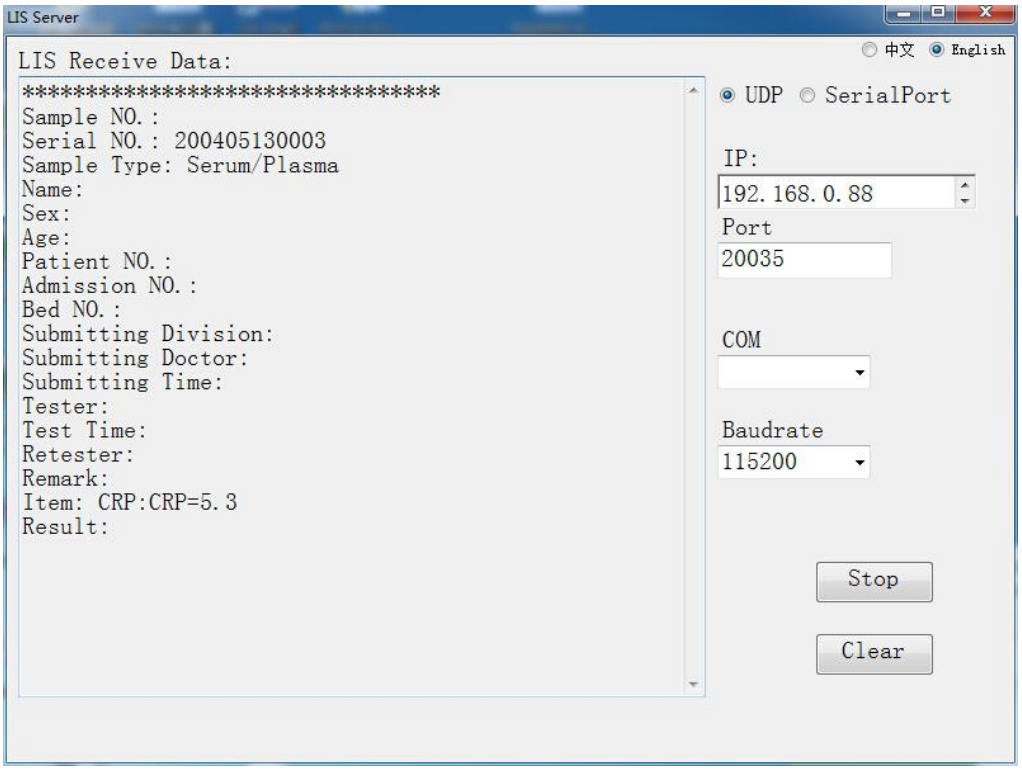


7) Click **"OK"**, return to IP address setting menu, then click **"OK"** to exit IP setting, run the software of ReceiveFromCRPTemp.exe after setting successful.

Note：**Finecare™ FIA Meter Plus** will send the current test results to computer automatically after test(in the case of LIS system is connected). Click "**Record**", selected historical record can also be sent by clicking "**Send**".

Run the software of ReceiveFromCRPTemp.exe, **Finecare™ FIA Meter Plus** will send the testing results as follows. (In an example of CRP, see **Fig. 6.13**)

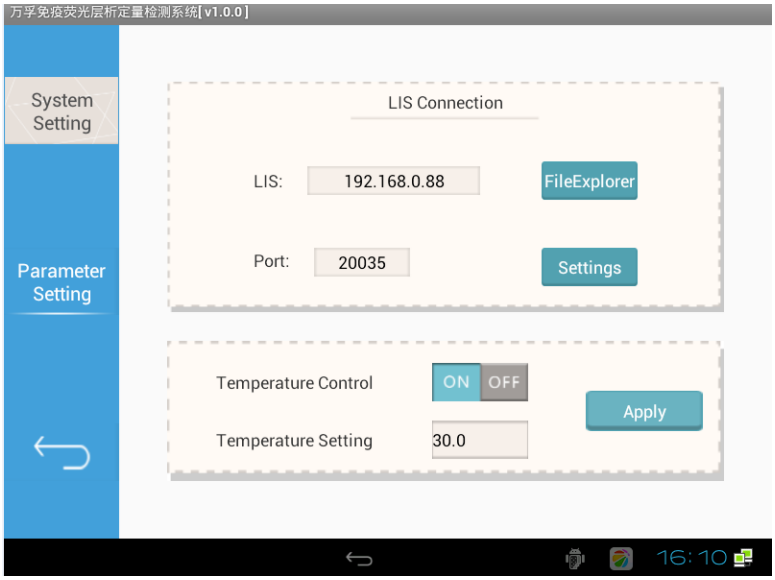
Fig. 6.13



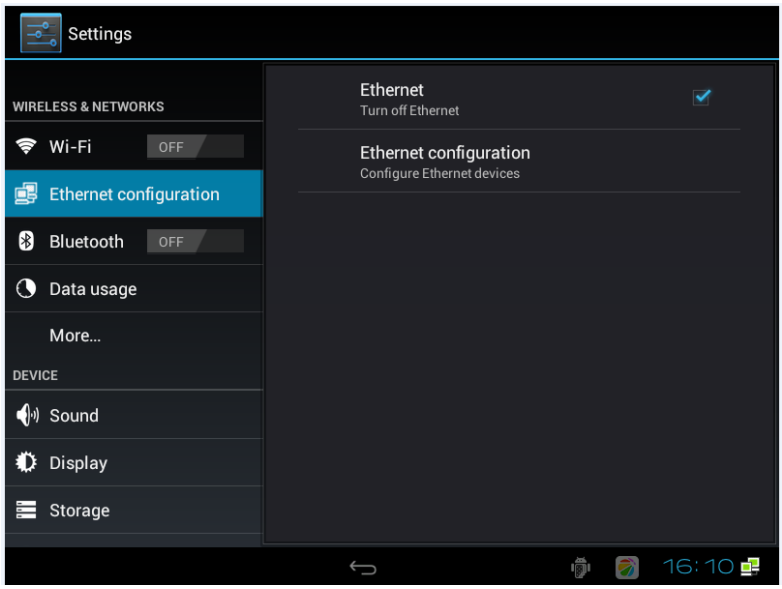
Ⅵ.1.2 Instrument terminal IP setup steps

1) Enter "System Settings" interface, click "Settings", input the corresponding IP address with computer :

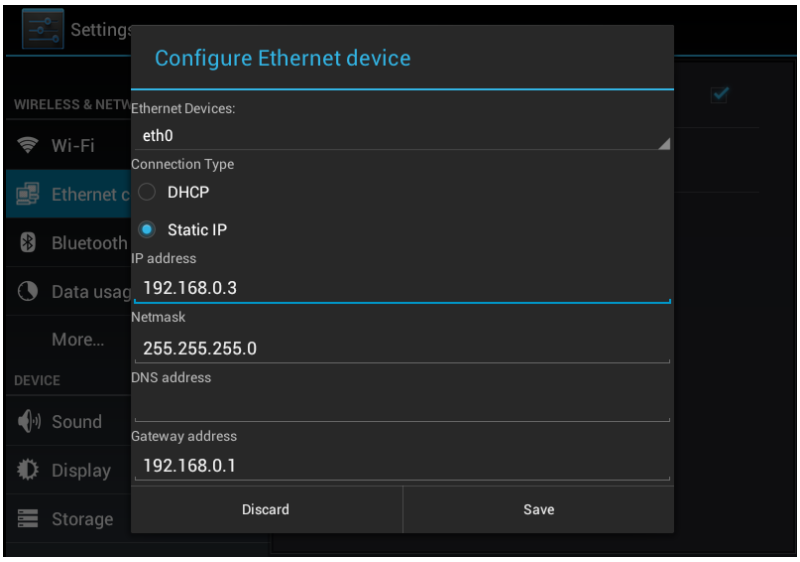
"192.168.0.88" .



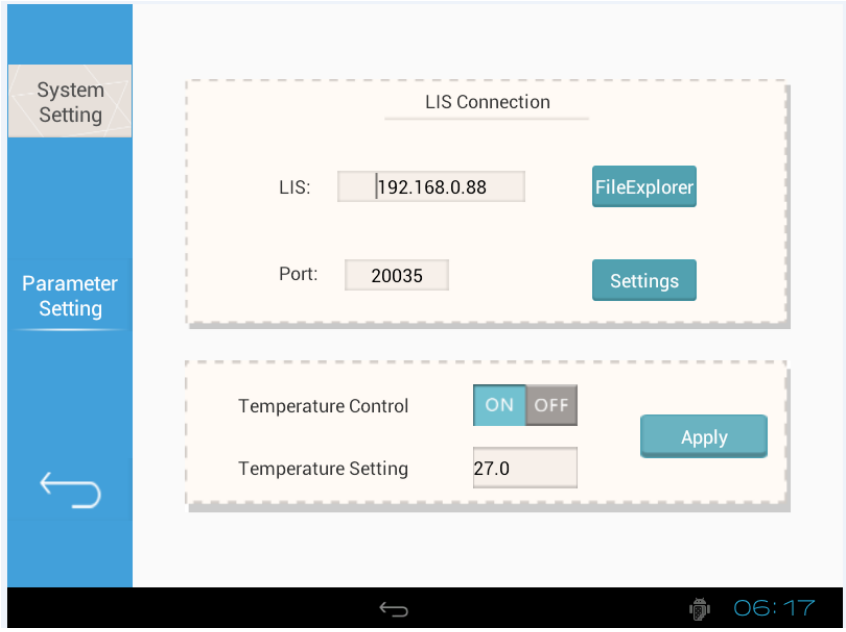
2) Choose “Ethernet configuration” to enter Network Settings interface.



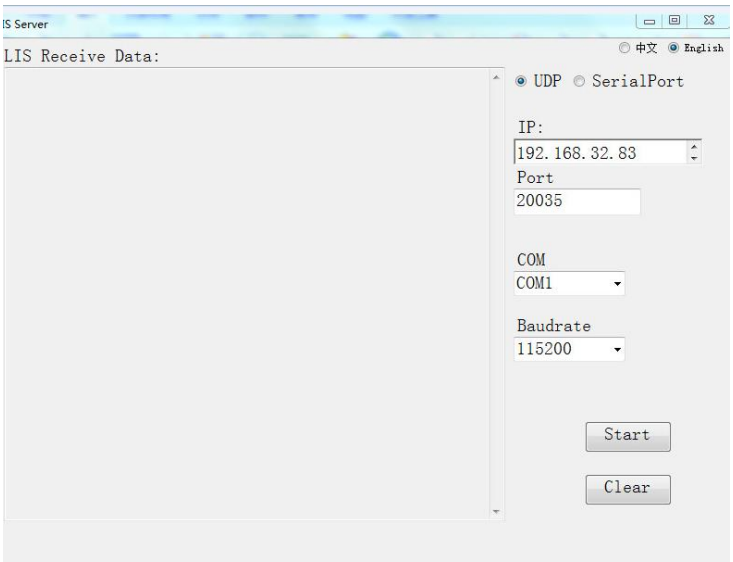
3) Click “Turn on Ethernet”, pop-up setting box where you can set the IP address of instrument terminal.



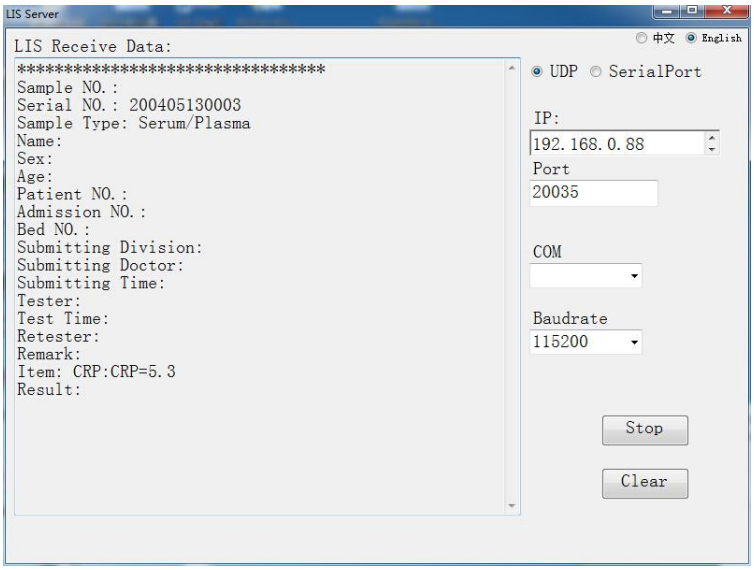
4)) Set the network port IP address of instrument for LIS transmission and ensure consistency with IP address of the computer terminal.



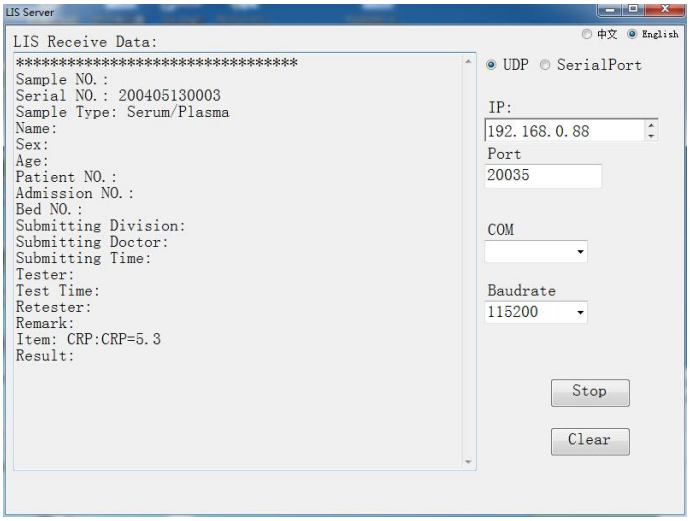
5) Turn on LIS data receiving software of the computer, select "network port" and click “Start”.
Note: LIS serial transmission function has been canceled, not verify here.



6) The instrument can test normally, the test results will be sent automatically to the PC reception software. If you did not input the details of test item, such as name and sample types, computer terminal will be processed as blank at this option.



7) Or check specific items and click “LIS” to upload the selected results to PC in the current result display interface.



Section Ⅶ Quality Control

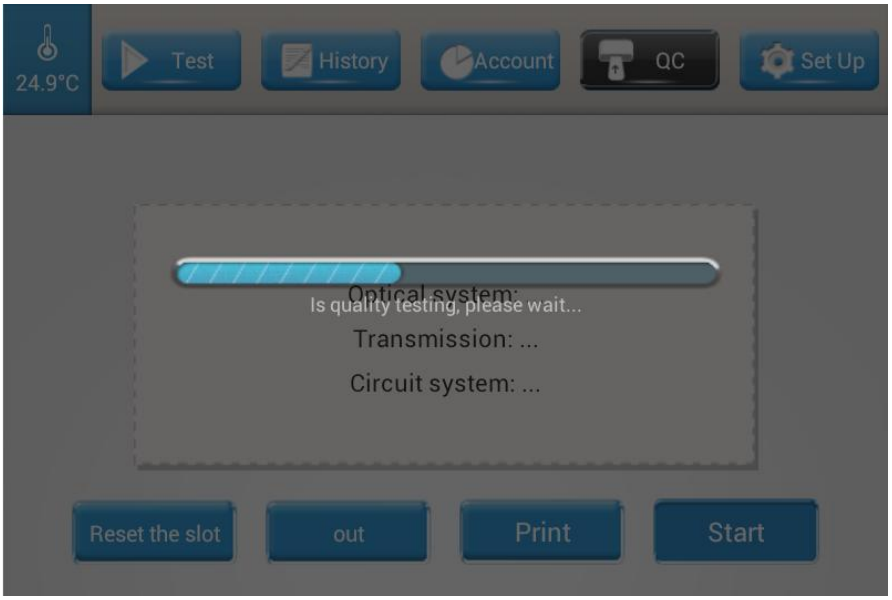
When the **Finecare™ FIA Meter Plus** is used for the first time or each time, user can use the Quality Control Card to check the internal systems of the meter (including Optical System, Transmission System, Circuit System), to ensure instrument is functioning properly before testing through quality inspection of instrument. Please follow the operation steps as below:

Step 1 Click "**QC**" on the top of the instrument to enter the quality control interface, it is used to make sure **Finecare™ FIA Meter Plus** to work properly.

Step2 Push the Quality Control Card all the way into the Test Device Holder until the card comes to a stop (Please click "**Out**" if Text device Holder did not come out.). Use only the push of your thumb or the index finger. Don't use excessive force on the Test Device, or it will result in a mechanical failure.

Step 3 Click "**Start**" to start quality inspection. (See **Fig. 7.1**)

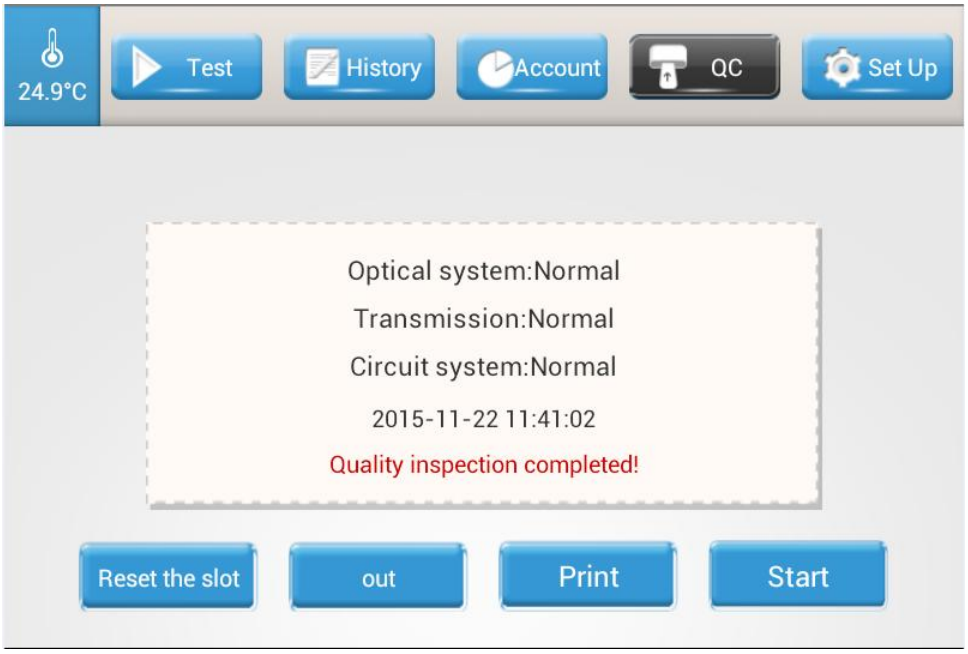
Fig. 7.1



Caution:
When Test Device Holder is not come out, click "Out", the Test Device Holder will move out of the main body of the instrument.)

Step4 Click "**Print**" after completing the quality inspection to print the quality inspection result. (See **Fig. 7.2**)

Fig. 7.2



Step 5 **C** lick “Reset the solt” if you do not need to use.

Section Ⅷ Warning, Precautions and Limitations

Caution in Handling the Test Device

- The test sample should be applied to the Sample Well B, as shown in **Fig. 2.8** of ***Section Ⅱ***.

- When inserting the Test Device into the **Test Device Holder**, orient the Test Device in such a way that the Sample Well B should be toward **Finecare™ FIA Meter Plus.** This direction is also marked with an arrow on the Test Device.

- Push the Test Device all the way into the Test Device Holder until the Test Device comes to a stop. Use only the push of your thumb or the index finger. Don't use excessive force on the Test Device, or it will result in a mechanical failure.

- The used Test Device should be treated as a potential bio-hazard and should be disposed of according to the standard procedures and relevant regulations observed by microbiological hazard materials. We recommended high temperature autoclaving by incineration.

- Gloves, masks and other protective measures should be used when handling potential infectious materials.

Caution in Use with the Finecare™ FIA Meter Plus

- Do not insert anything other than a Test Device provided by the manufacturer into the **Test Device Holder**.

- Do not spill any liquid on **Finecare™ FIA Meter Plus.** This may disable the system.

- Do not drop or crash **Finecare™ FIA Meter Plus.**

- Do not subject **Finecare™ FIA Meter Plus** to mechanical shocks heavier than light taps with hands.

- Do not dismantle **Finecare™ FIA Meter Plus** without written authorization from Guangzhou Wondfo Biotech Co., Ltd. or its representative.

- Do not place heavy objects on **Finecare™ FIA Meter Plus.** This may damage the optical alignment to result in degraded performance or in mechanical damage.

- The mains outlet for **Finecare™ FIA Meter Plus** shall be located in a user accessible position for easy unplugging in emergency.

- Please make sure power outlet properly grounded. Improper grounding may cause electrical shock

and systematic damage.

- Use only the **Power Adapte**r (including the power cord) provided with **Finecare™ FIA Meter Plus.**

- Use **Finecare™ FIA Meter Plus** only in environment as specified in **Section** Ⅲ.

- **Finecare™ FIA Meter Plus** malfunction due to maintenance, processing or transporting damage, please contact customer service representative.

- Use the test result as a pre-screening guide. The result should be interpreted by trained medical personnel only.

The **Finecare™ FIA Meter Plus** shall be operated by professional person.

- If the **Finecare™ FIA Meter Plus** is used in a manner not specified by the manufacturer, the protection provided by the instrument may be impaired.

***Cautions** should be taken to use **Finecare™ FIA Meter Plus** in Settings meeting the requirements specified in **Section** Ⅲ of this manual and in observance of the warning in **Section** Ⅶ. Otherwise, the built-in safety features of **Finecare™ FIA Meter Plus** may be compromised to present severe electrical, mechanical or biological hazards to the user.*

Section Ⅸ Service, Maintenance and Disposal

If any service or maintenance were to be required, **Finecare™ FIA Meter Plus** should be sent to Guangzhou Wondfo Biotech Co., Ltd. No Special maintenance other than paper replacement and periodic cleaning is required for the instrument. Occasional cleansing of the exterior with a mild-dry cloth is sufficient to assure the instrument operation. The instrument shall be separated from mains supply prior to cleaning.

External cleaning

Use a damp cloth with 0.5% bleach, 70% isopropyl alcohol or 70% ethanol to clean the external surface of instrument. To prevent damages of screen or other surface device, strong bleach (bleach concentration more than 0.5%), oxidizing substances and solvents are prohibited.

Decontamination

If the instrument needs maintenance or replacement after clinical use, decontamination and **disinfection** before repacking and transporting should be done. Use disinfectant (e.g. bleach concentration lower than 0.1%) and a cloth to scrub external surface of instrument completely. To prevent damages of inner devices, spray-washing or clean any internal parts and inner surface with disinfectant is prohibited.

Caution:
Instrument does not contain any serviceable components by operator. To avoid electric shock, regular maintenance must be executed by authorized technical service personnel.

Maintenance

In addition to regularly cleaning or replace paper, **Finecare™ FIA Meter Plus** generally does not require special maintenance, with a soft, dry cloth to wipe the external surface of the instrument to ensure the normal operation of **Finecare™ FIA Meter Plus**.

Power Supply

Use only the AC/DC power supply provided with the instrument. In case the power adapter needs replacement, you should contact **Guangzhou Wondfo Biotech Co., Ltd.** or its designated representative. **Guangzhou Wondfo Biotech Co., Ltd.** CE certified Class 2 power supplies, rated at +12V 5A.

Return Procedure

If **Finecare™ FIA Meter Plus** happened to malfunction, please contact **Guangzhou Wondfo Biotech Co., Ltd.** or the local distributor at first. If it were determined that the unit will be returned to the manufacturer, a return authorization number will be issued, then **Guangzhou Wondfo Biotech Co., Ltd.** will send a **Finecare™ FIA Meter Plus** for replacing**.** The user is expected to utilize the packaging supplies accompanying the replacement to ship the malfunctioning unit in. Verify the return authorization number on the package and send the unit to **Guangzhou Wondfo Biotech Co., Ltd.** upon receiving the replacement **Finecare™ FIA Meter Plus.**

Transporting and storing

The original shipping container should be used to ship or transport **Finecare™ FIA Meter Plus.** The original shipping container is also recommended for storing **Finecare™ FIA Meter Plus** over an extended period. When transporting or storing **Finecare™ FIA Meter Plus,** keep it dry in upright position and protect it from sun, mechanical shocks, transportation requirements according to an order under the contract.

Finecare™ FIA Meter Plus should be stored in a well-ventilated room that has storage temperature of -10℃~50℃, relative humidity is less than 85% and no corrosive gas.

Disposal

Any spent Test Device should be disposed according to local ordinances regarding the disposal of bio-hazardous materials.

WEEE (Waste Electrical and Electronic Instrument)

This symbol for the marking of electrical and electronics devices is according to the Directive 2002/96/EC. The **Finecare™ FIA Meter Plus**, accessories and the packaging have to be disposed of waste correctly at the end of usage. Please follow Local Ordinances or Regulations for disposal.

Section X Trouble Shooting

Phenomena	Probable Cause	Recommended Measures
Finecare™ FIA Meter Plus does not response	Power Failure	Check the Power Plug.
	Poor Connection between the Power Adapter and the Cord.	Pull out the Cord and Re-connect firmly.
	Main switch is off.	Turn on the main switch.
	Power Adapter is broken.	Call Customer Service.
Run finished but no result returned	Excessive Computation Load	Wait till the computation is finished.
	Computational Abnormality	Turn off the power and start fresh.
Test Device Holder does not come out	Mechanical Failure (Buzzing Noise)	Call Customer Service.
	Software Abnormality.	Turn off the power and start fresh.
Clicking noise when the Test Device Holder moves	Normal Mechanical Setting	No User Action required.
Display Screen does not function properly	Electrostatic Influence	Discharge any static electricity, reboot the system
	Electric Circuit Failure.	Call Customer Service.
Message: Reset reagent and test again!	Test device or ID chip was not inserted.	Insert the test device or ID chip.
Message: Insert the corresponding ID chip	ID chip is not inserted or the lot of Test Device is incompatible.	Insert the right ID chip.
Message: "Optical System Fault", "Transmission System Fault", and/or "Circuit System Fault"	Internal hardware failure	Please Contact Wondfo Customer Service Center.

Section XI Contact Information

Guangzhou Wondfo Biotech Co., Ltd. ’s expressed and implied warranties are conditioned upon full observance of manufacturer’s published direction with respect to the use of **Guangzhou Wondfo Biotech Co., Ltd.** ’s products. Under no circumstance whatsoever shall **Guangzhou Wondfo Biotech Co., Ltd.** be held liable for any indirect or consequential damages.

For technical assistance, call or send e-mail to us:

Manufacturer

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EC REP

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